Strengthening the Reframed OOD Paper for ICLR 2027

A prioritized technical, empirical, and writing roadmap

From MM++ toward a detector-agnostic layer-selection and feature-fusion framework

Executive recommendation

The reframing and renaming are necessary, but the highest-leverage improvement is to convert the rebuttal's strongest diagnosis into a sharper algorithmic contribution: **complementarity-aware layer selection**. The current entropy-drop rule identifies useful transitions, but it can select a layer that is highly redundant with the terminal anchor. An ID-only novelty term can explicitly reward layers that add information not already represented by the active set.

Priority	Action	Expected impact	Effort
1	Complementarity-aware selector	Very high	Medium
2	Evidence-supported block-diagonal scoring	High	Low–medium
3	ID-only fallback when fusion is unhelpful	High	Medium
4	Modular multi-backend evaluation	High	Medium
5	Selection-focused theory and stability	Medium–high	Medium

1 Add complementarity-aware layer selection

This is the strongest opportunity. The current entropy-drop rule selects `block_11` at $K = 3$, even though it is almost redundant with the final `norm` representation. The existing analysis already shows:

- canonical correlation $= 0.976$;
- only 3.3% residual variance is new;
- replacing it with `block_06` raises AUROC from 78.58 to 83.41 at the same fused dimension.

Turn that observation into the revised selector. After choosing a layer set $\mathcal{S}$, define the conditional residual covariance of candidate layer l:

$$\Sigma_{l|\mathcal{S}} = \Sigma_{ll} - \Sigma_{l\mathcal{S}}\Sigma_{\mathcal{S}\mathcal{S}}^{-1}\Sigma_{\mathcal{S}l}. \tag{1}$$

Then define an ID-only novelty score:

$$\nu(l \mid \mathcal{S}) = \frac{\mathrm{tr}(\Sigma_{l|\mathcal{S}})}{\mathrm{tr}(\Sigma_{ll})}. \tag{2}$$

Combine it with the entropy drop:

$$q_l = \max(\Delta_l, 0)\,\nu(l \mid \mathcal{S}). \tag{3}$$

This selector asks two questions:

1. Is the layer located at an informative spectral transition?
2. Does it add information not already represented by the selected layers?

Why this materially improves the paper

A duplicate layer has $\nu = 0$; an independent layer has high residual novelty. The method remains training-free and ID-only, directly repairs the $K = 3$ failure, and turns a rebuttal diagnosis into a principled algorithmic improvement rather than a post-hoc explanation.

2 Use the covariance estimator supported by the evidence

The off-diagonal ablation shows:

Estimator	Mean AUROC	Mean FPR95
Full covariance	82.36	53.43
Block diagonal	82.54	53.04
Within-class permutation	82.49	53.24

The full cross-layer covariance is not helping in the tested setting. For the revised Mahalanobis backend, consider the block-diagonal score

$$S(x) = -\min_c \sum_{l \in \mathcal{S}} d_{c,l}(x), \tag{4}$$

with independently regularized layer covariances where appropriate.

Advantages:

- easier covariance estimation;
- lower memory and inversion cost;
- natural support for heterogeneous layer widths;
- preservation of a shared class hypothesis across selected layers.

Run the missing comparison

$$-\min_c \sum_l d_{c,l}(x) \quad \text{versus} \quad \sum_l -\min_c d_{c,l}(x). \tag{5}$$

If the first wins, the paper retains a clean detector-specific contribution: one class must jointly explain all selected depths, even without off-diagonal covariance. Repeat the full-versus-block-diagonal test on Swin and ConvNeXt; if the result persists, simplify the method rather than retaining unsupported complexity.

3 Add an ID-only fallback rule

The 33-checkpoint evaluation shows that fusion does not help every training recipe, particularly some DeiT3 and SigLIP variants. A stronger framework should know when *not* to fuse.

Possible ID-only signals include:

- maximum entropy-drop magnitude;
- residual novelty of the chosen layer;
- selection stability across bootstrap calibration subsets;
- CKA or canonical correlation with the final representation;
- margin between the highest and second-highest selection scores.

A practical rule is:

> Use multilayer fusion only when the selected layer has residual novelty above an ID-derived stability threshold; otherwise, use the final representation alone.

Evaluate the rule across all 33 checkpoints and report wins, ties, losses, worst-case degradation, and average improvement before and after fallback. If it removes the major SigLIP/DeiT3 failures, it becomes a meaningful practical contribution.

4 Define the framework modularly

The revised method should have three explicit modules:

$$\text{ID spectral selector} \longrightarrow \text{normalized feature fusion} \longrightarrow \text{downstream detector}. \tag{6}$$

Supported instantiations should be named clearly:

- LSF-Mahalanobis;
- LSF-KNN;
- LSF-KPCA.

Do not list Energy, SCALE, or NNGuide as downstream instantiations unless the selected fused representation is actually passed into those methods.

A clean core ablation is:

Selector	Fusion	Backend
Final-only	None	Maha / KNN / KPCA
Random layer	Concatenation	Maha / KNN / KPCA
Last intermediate	Concatenation	Maha / KNN / KPCA
Entropy drop	Concatenation	Maha / KNN / KPCA
Complementarity-aware entropy	Concatenation	Maha / KNN / KPCA
Oracle	Concatenation	Diagnostic only

This establishes both detector generality and the value of the selector.

5 Strengthen theory around selection, not Gaussianity

Gaussian-compatibility diagnostics are useful validation, but the general framework should not depend on a Gaussian backend. More valuable formal results would address the selector itself.

5.1 Scale invariance

Prove that independently scaling any selected layer by a positive scalar does not alter its normalized representation.

5.2 Selection stability

Develop a bound of the form

$$\|\widehat{\Sigma}_l - \Sigma_l\|_2 \leq \epsilon_l \quad \Longrightarrow \quad \left|\widehat{H}_l - H_l\right| \leq f(\epsilon_l, D_l), \tag{7}$$

then state a margin condition under which the top-K layer set remains unchanged.

5.3 Complementarity property

For the conditional residual score, establish that exact linear redundancy implies zero residual covariance, orthogonal additional information yields nonzero novelty, and the score is invariant to suitable reparameterizations within the selected span.

5.4 Shrinkage stability

Retain the precision perturbation bound and positive-eigenvalue guarantee as theory for the Mahalanobis instantiation, not as the theory of the general framework.

6 Explain when and why the framework works

Use the existing 33 checkpoints to correlate OOD gains with:

- entropy-drop magnitude;
- residual novelty;
- CKA with the final layer;

- effective-rank increase;
- model training recipe;
- architecture family.

The revised paper can answer a broader question:

> What ID-only properties predict whether intermediate-layer fusion will help?

A particularly strong result would show that residual novelty predicts improvement and can be used for fallback or adaptive K.

7 Make the evaluation visibly detector-agnostic

At minimum, report Mahalanobis, KNN, and KPCA across:

- ViT-B/16, Swin-T, and ConvNeXt-T;
- ImageNet-1K and ImageNet-LT;
- near-OOD and semantic far-OOD groups separately.

Add paired bootstrap confidence intervals, calibration-subset variability, selected layers and dimensions, runtime and memory, and all meaningful failure cases. A non-ImageNet setting can improve generality, but it is lower priority than fixing the selector and failure handling.

8 Make reproducibility unusually strong

ICLR encourages clear reproducibility documentation and supplementary source code. Include:

- an anonymized repository at submission;
- exact commands for each main table;
- checkpoint identifiers;
- layer hooks and pooling rules;
- cached-feature format;
- bootstrap scripts;
- baseline configurations;
- one end-to-end script reproducing the principal table.

9 Rewrite the claims completely

Remove or narrow the following:

- "fully unsupervised";
- "universal robustness";

- “off-diagonal cross-layer dependencies drive the gains”;
- “every entropy drop is semantic compression”;
- “X-Maha necessarily requires fine-tuning”;
- “Gaussianity is validated exactly.”

A stronger and more accurate central claim is:

Recommended central claim

ID spectral geometry can identify a sparse set of complementary intermediate representations that improves multiple post-hoc feature-based OOD detectors, without OOD data or gradient adaptation.

10 Prioritized execution plan

The official ICLR 2027 paper deadline is September 25, 2026 (AOE). The highest-value order is:

1. Implement complementarity-aware selection.
2. Run old-versus-new selectors across the 33 checkpoints.
3. Test shared-class block-diagonal scoring.
4. Add the ID-only fallback rule.
5. Rebuild the paper and main tables around multiple backends.
6. Add selection-stability theory.
7. Finalize code and reproducibility materials.

11 What most improves the acceptance probability

Renaming improves clarity. Reframing plus the current experiments makes the paper credible. But reframing plus a complementarity-aware selector makes it a meaningfully stronger algorithmic contribution. It converts the most important rebuttal diagnosis—the redundant $K = 3$ layer—into a principled improvement.

Bottom line

The clearest path from a competitive resubmission to a stronger originality case is: **general framework + complementarity-aware selector + ID-only fallback + evidence-supported covariance backend**.

Official sources. ICLR 2027 reviewer criteria, author/reproducibility guidance, and submission dates are available from the official ICLR pages: Reviewer Guidelines, Author Guidelines, and Call for Papers.